PriceWise | EDA summary

Recent product selection and evidence for the pricing recommendation system

Status: The new notebook is prepared for your transaction CSVs. The actual CSV files were not attached; no new product rankings, revenues or pricing patterns have been computed from client data. Validation uses a separate synthetic fixture, not business evidence.

What the new EDA does

Analysis	What was done and why
Recent product portfolio	Select up to 30 product IDs by shipped-invoice transaction count in the latest six calendar months. Show each product's family, revenue, order units and weighted ASP. Compare coverage for 20, 25 and 30 products to support the initial portfolio decision.
Family portfolio	Select the five highest-transaction resolved families and up to six products per family. Report missing slots, portfolio overlap and conflicting family assignments so the selection can be repeated and reviewed.
Commercial and price patterns	Compute monthly and rolling 30/90-day metrics, normalized price tiers, tier mismatch flags, recent unit movement and observed .99 ASP endings. Use heatmaps and small multiples to make individual products readable.
Recommendation readiness	Flag sparse recent activity, stale products, invalid price inputs and family ambiguity. Show observed sales gaps and compare family ASP with product ASP before relying on a family benchmark.

Population and metric choices

Stage 1 population: Enterprise shipped invoices, following the updated business scope. "For everyone" is interpreted as a team deliverable; the segment setting can be changed if all segments were intended. Transaction count means input row count because there is no verified invoice-line identifier. Repeated rows remain in the data; each receives a source-file/source-row ID.

Time: The default six-month window ends at the latest valid shipment date within each country. This is data-relative rather than wall-clock-relative; an explicit reporting-date override is available. Shipment date is a stated proxy for invoice date. Missing dates are audited, not replaced by fiscal filenames. Fiscal Q1 starts in April; partial months are labeled.

Money: Weighted ASP = total revenue / total order units. Margin rate = total margin / total revenue. Country results are separate until the currency basis is confirmed. The notebook audits differences between revenue fields and checks margin against revenue minus cost. Tax is not added again.

What changed and what comes next

Why the original EDA needed a fresh structure

The original notebook provides useful broad profiling, but its saved results cover all segments and statuses. Its saved merge contains 11,738,282 rows across 14 CSV files; its duplicate check reports 1,358,057 repeated rows (11.57%). These are old-notebook outputs, not freshly recomputed findings.

All-time revenue leaders do not answer the recent-transaction selection question. Product-name grouping can merge distinct IDs. Record-based returns are different from returned units / shipped units. Tier labels with different spacing split equivalent buckets, and fiscal filenames can disagree with the embedded period or shipment date. The new EDA addresses these issues explicitly rather than recycling the original portfolio.

How the EDA supports recommendation design

System component	EDA evidence and decision
Baseline price	Current product 30-day ASP, family 90-day ASP and previous non-overlapping product 90-day ASP are shown together. The scope's illustrative 0.5/0.3/0.2 baseline is a diagnostic; missing components and unlike family prices remain flagged.
Objective signals	Revenue importance, margin, recent unit movement and weak-sales indicators inform transparent stakeholder objectives. A unit decline does not prove that a price cut will help; inventory is needed to establish clearance pressure.
Guardrails	Tier normalization and mismatches identify mappings needing approval. Operational rules still need validated tier boundaries, a margin floor, monthly reference prices and the previous day's suggestion. .99 rounding must be followed by another guardrail check.
Explainability and evaluation	Export product-level metrics, review reasons and selection membership. Future back-tests should measure coverage, stability and guardrail frequency. Historical price correlations alone cannot establish revenue or margin uplift.

How to use the deliverable

Open **PriceWise_Recent_Products_EDA.ipynb**, set DATA_DIR to the folder containing synthesized_FY*.csv, review the configuration and choose Run All. The default input path matches the original notebook. Install pandas, numpy, matplotlib and seaborn if needed. The run writes country-specific selection tables, rolling metrics, monthly patterns, pricing flags, charts and a computed FINDINGS.md under pricewise_eda_results.

Confirm with stakeholders

Confirm the revenue/cost/margin definitions and currency basis; the latest complete reporting date and shipment-date proxy; whether Shipping, Services and Options should remain eligible; product-family mappings; and recent-data thresholds. The requested rankings currently retain all eligible product types and attach review flags rather than silently excluding them.

Limits: Six months can reveal recent patterns but cannot validate recurring annual seasonality. Family averages mix products and are not automatic fallback prices. Return ratios are unlinked same-window flows, not cohort probabilities. The EDA does not estimate causal price elasticity, optimize weights, invent inventory or recommend unsupported prices.

Sources reviewed: EDA.ipynb; Business Scope(3).pdf; 0925_PriceWise_Business_Scope.pptx; 0925 Client Meeting Notes.pdf; latest team selection instructions.